

Myosin MgADP release rate decreases at longer sarcomere length to prolong myosin attachment time in skinned rat myocardium

Bertrand C. W. Tanner, Jason J. Breithaupt, and Peter O. Awinda

Department of Integrative Physiology and Neuroscience, Washington State University, Pullman, Washington

Submitted 14 July 2015; accepted in final form 5 October 2015

Tanner BC, Breithaupt JJ, Awinda PO. Myosin MgADP release rate decreases at longer sarcomere length to prolong myosin attachment time in skinned rat myocardium. *Am J Physiol Heart Circ Physiol* 309: H2087–H2097, 2015. First published October 14, 2015; doi:10.1152/ajpheart.00555.2015.—Cardiac contractility increases as sarcomere length increases, suggesting that intrinsic molecular mechanisms underlie the Frank-Starling relationship to confer increased cardiac output with greater ventricular filling. The capacity of myosin to bind with actin and generate force in a muscle cell is Ca^{2+} regulated by thin-filament proteins and spatially regulated by sarcomere length as thick-to-thin filament overlap varies. One mechanism underlying greater cardiac contractility as sarcomere length increases could involve longer myosin attachment time (t_{on}) due to slowed myosin kinetics at longer sarcomere length. To test this idea, we used stochastic length-perturbation analysis in skinned rat papillary muscle strips to measure t_{on} as [MgATP] varied (0.05–5 mM) at 1.9 and 2.2 μm sarcomere lengths. From this t_{on} -MgATP relationship, we calculated cross-bridge MgADP release rate and MgATP binding rates. As MgATP increased, t_{on} decreased for both sarcomere lengths, but t_{on} was roughly 70% longer for 2.2 vs. 1.9 μm sarcomere length at maximally activated conditions. These t_{on} differences were driven by a slower MgADP release rate at 2.2 μm sarcomere length (41 ± 3 vs. $74 \pm 7 \text{ s}^{-1}$), since MgATP binding rate was not different between the two sarcomere lengths. At submaximal activation levels near the pCa_{50} value of the tension-pCa relationship for each sarcomere length, length-dependent increases in t_{on} were roughly 15% longer for 2.2 vs. 1.9 μm sarcomere length. These changes in cross-bridge kinetics could amplify cooperative cross-bridge contributions to force production and thin-filament activation at longer sarcomere length and suggest that length-dependent changes in myosin MgADP release rate may contribute to the Frank-Starling relationship in the heart.

cross-bridge kinetics; myosin attachment time; length-dependent activation; frank-starling relationship

NEW & NOTEWORTHY

Cardiac force production arises from myosin cross-bridge binding, which is regulated by Ca^{2+} -activation and sarcomere length throughout a heartbeat. We show that myosin MgADP release rate slows with sarcomere length to prolong myosin attachment time, which enhances force production at longer sarcomere length and may contribute to the Frank-Starling relationship.

THE ACTIN-MYOSIN CROSS-BRIDGE interaction transforms chemical energy from ATP hydrolysis into mechanical energy to provide the force and shortening that is required for the heart to pump blood throughout the body. The capacity of myosin to generate force is regulated by thin-filament proteins and sarcomere

length, both of which dictate the number of potential actin-myosin cross-bridge interactions during contraction. At the organ level, the Frank-Starling law of the heart links increased contractility with increased ventricular filling to dynamically accommodate contractile demand by modulating contractility on a beat-to-beat basis. Although considerable efforts have investigated the molecular underpinnings of this important physiological response over the past 40–50 years (3, 10, 18, 26, 35, 36), we still lack a detailed understanding of the molecular mechanisms that combine to facilitate this response.

A number of myofilament mechanisms could contribute to the Frank-Starling relationship, such as increased Ca^{2+} -sensitivity of thin-filament activation (13, 14, 38, 41, 49, 56, 62), reduced thick-to-thin filament surface distance (7, 15, 19, 31, 37, 44), or varied myosin kinetics with sarcomere length (2, 17, 39, 47, 57). These mechanisms contribute to the consistent observation that Ca^{2+} -sensitivity of the force-pCa relationship (where $\text{pCa} = -\text{Log}_{10}[\text{Ca}^{2+}]$) increases as sarcomere length increases. In cardiac muscle, this is thought to arise from a couple forms of cooperativity that enhance thin-filament activation: increased Ca^{2+} -affinity or opening of troponin-C as sarcomere length increases (27, 28, 41, 56, 65) and increased cross-bridge binding to stabilize or amplify thin-filament activation as sarcomere length increases (4, 16, 46). The cross-bridge contributions to thin-filament activation could become stronger as thick-to-thin filament spacing decreases with increasing sarcomere length (31, 37, 43, 56), thereby enhancing the probability of cross-bridge binding at longer sarcomere length. Consistently, cross-bridge contributions to thin-filament activation could be enhanced by slower cross-bridge cycling rates at longer sarcomere length (1, 17, 24, 25, 39, 47, 53, 57), although others have reported that sarcomere length does not affect cross-bridge kinetics (11, 23, 66). These length-dependent mechanisms of thin-filament activation and cross-bridge binding likely combine to influence contractility throughout a heartbeat.

To address the hypothesis that cross-bridge contributions to the Frank-Starling relationship may arise from length-dependent effects on fundamental steps of the cross-bridge cycle, we used stochastic length-perturbation analysis to assess myosin kinetics at 1.9 and 2.2 μm sarcomere lengths in skinned strips of rat papillary muscle. This method allows us to measure the stress-strain response across a broad frequency range in a muscle fiber during isometric contraction using a short burst of noise (60). Frequency-dependent shifts in this stress-strain relationship that vary with sarcomere length arise from varied cross-bridge kinetics at fundamental steps in the cross-bridge cycle, such as phosphate release, MgADP release, or MgATP binding rates. We find that myosin MgADP release rate slows with increased sarcomere length to prolong the average myosin attachment duration (t_{on}) in skinned myocardial strips, which

Address for reprint requests and other correspondence: B. C. W. Tanner, Rm. 255 Vet/Biomed Research Bldg., 1815 Ferdinand's Lane, Dept. of Integrative Physiology and Neuroscience, Pullman, WA 99164-7620 (e-mail: btanner@vetmed.wsu.edu).

could be an intrinsic mechanism that enhances contractility and thin-filament activation at longer sarcomere lengths to facilitate the Frank-Starling response.

MATERIALS AND METHODS

Animal models. All procedures were approved by the Institutional Animal Care and Use Committee at Washington State University and complied with the *Guide for the Use and Care of Laboratory Animals* published by the National Institutes of Health. Adult male Sprague-Dawley rats (19–24 wk) were acquired from Simonsen Laboratories (Gilroy, CA). Rats were anesthetized by isoflurane inhalation (3% volume in 95% O₂-5% CO₂ flowing at 2 l/min), following which hearts were immediately excised and placed in pre-oxygenated Krebs-HEPES solution on ice.

Solutions for skinned myocardial strips. Chemicals and reagents were obtained from Sigma (St. Louis, MO) unless otherwise noted. Solution concentrations were formulated by solving equations describing ionic equilibria according to Godt and Lindley (20), with all concentrations reported in mM. Krebs-HEPES solution contained 137 Na⁺, 153 Cl⁻, 5.4 K⁺, 0.2 Ca²⁺, 1.3 Mg²⁺, 10 Glucose, 10 HEPES, and 30 BDM (pH 7.4). Relaxing solution contained pCa 8.0, 5 EGTA, 5 MgATP, 1 Mg²⁺, 0.3 P_i, 35 phosphocreatine, 300 U/ml creatine kinase, and 200 ionic strength (pH 7.0). The activating solution was the same as relaxing with pCa 4.8. The rigor solution was the same as activating solution without MgATP. The skinning solution was the same as relaxing without creatine kinase, with 1% Triton X-100 wt/vol and 50% glycerol wt/vol. The storage solution was the same as skinning without Triton, with 10 µg/ml leupeptin. The alkaline phosphatase (AP) solution was the same as relaxing solution with 6 U/ml recombinant AP from E-coli (P-4252).

Skinned myocardial strips. Left ventricular papillary muscles were dissected from five hearts and pared down to thin strips (~180 µm in diameter and 700 µm long). Strips were skinned overnight at 4°C and stored at -20°C for up to 1 wk. Aluminum T-clips were attached to the end of each strip, and strips were incubated for 10 min in AP solution at room temperature immediately before mechanical analysis. Strips were mounted between a piezoelectric motor (P841.40; Physik Instrumente, Auburn, MA) and a strain gauge (AE801; Kronex, Walnut Creek, CA), lowered into a 30 µl droplet of relaxing solution, and stretched to 1.9 or 2.2 µm sarcomere length measured by digital Fourier Transform (IonOptix Corp, Milton, MA). Strips were calcium activated from pCa 8.0 to pCa 4.8, following which rigor solution was exchanged for activating solution to titrate [MgATP] to <0.001 mM. Temperature was maintained at 17°C throughout all experiments. At least two strips from each heart underwent this experimental protocol at each sarcomere length, and thirteen strips were included for these analyses at 1.9 and 2.2 µm sarcomere length.

Dynamic mechanical analysis. Stochastic length perturbations were applied for a period of 40 s similar to previously described measurements (60), using an amplitude distribution with a standard deviation of 0.05% muscle length over the frequency range 0.5–250 Hz. Elastic and viscous moduli, $E(\omega)$ and $V(\omega)$, were measured as a function of angular frequency (ω) from the in-phase and out-of-phase portions of the tension response to the stochastic length perturbation. The complex modulus, $Y(\omega)$, was defined as $E(\omega) + iV(\omega)$, where $i = \sqrt{-1}$. Fitting Eq. 1 to the entire frequency range of moduli values provided estimates of six model parameters (A , k , B , $2\pi b$, C , $2\pi c$): $Y(\omega) = A(i\omega)k - B[(i\omega)/(2\pi b + i\omega)] + C[(i\omega)/(2\pi c + i\omega)]$ (Eq. 1).

The A-term in Eq. 1 reflects the viscoelastic mechanical response of passive, structural elements in the muscle and does not reflect any enzymatic behavior. The parameter A represents the combined mechanical stress of the fiber, and the parameter k describes the viscoelasticity of these passive elements, where $k = 0$ represents a purely elastic response and $k = 1$ is a purely viscous response (48, 51). The B- and C-terms in Eq. 1 reflect enzymatic cross-bridge cycling behavior that produce frequency-dependent shifts in the viscoelastic

mechanical response during Ca²⁺-activated contraction, which characterize work-producing and work-absorbing processes, respectively (6, 32, 50, 68). From a physiological standpoint, therefore, the B-process stems from the muscle generating work throughout a length-perturbation (i.e., producing positive mechanical energy) and is associated with cross-bridge attachment or recruitment events that enhance force production and/or muscle shortening (6). Consistently, the C-process stems from muscle absorbing mechanical energy throughout a length-perturbation and is associated with cross-bridge detachment events that diminish force production (50). The parameters B and C represent the mechanical stress from the cross-bridges (i.e., number of cross-bridges formed $\times$ their mean stiffness), and the rate parameters $2\pi b$ and $2\pi c$ reflect cross-bridge kinetics that are sensitive to biochemical perturbations affecting enzymatic activity, such as [MgATP], [MgADP], or [P_i] (42). Molecular processes contributing to cross-bridge attachment or force generation underlie the cross-bridge attachment rate, $2\pi b$, and processes contributing to cross-bridge detachment or force decay underlie the cross-bridge detachment rate, $2\pi c$. Cross-bridge detachment rate is inversely related to the average myosin attachment time, $t_{on} = (2\pi c)^{-1}$ (50).

Stochastic system analysis provided a portrait of cross-bridge kinetics as a function of [MgATP] under maximally activated conditions. Assuming that the average myosin t_{on} in a skinned strip under these conditions is composed of the myosin-ADP and myosin-rigor states, the cross-bridge detachment rate, $2\pi c$, can be described by $2\pi c = k_{ADP}[MgATP]/\{(k_{ADP}/k_{+ATP}) + [MgATP]\}$ (Eq. 2).

As explained in detail by Tyska and Warshaw (63) and implemented in our previous publications (61, 64), fitting the $2\pi c$ -MgATP relationship to Eq. 2 allows a calculation of 1) k_{ADP} , which represents cross-bridge MgADP release rate and the asymptotic, maximal myosin detachment rate in s⁻¹ at saturating [MgATP]; 2) k_{+ATP} , which represents the second-order cross-bridge MgATP binding rate in M⁻¹s⁻¹; and 3) the ratio of k_{ADP}/k_{+ATP} , which represents the concentration of MgATP producing half the maximal myosin detachment rate ($[MgATP]_{50} = k_{ADP}/k_{+ATP}$).

Statistical analysis. All values are means $\pm$ SE. Constrained nonlinear least squares fitting of Eqs. 1 and 2 to the data was performed using sequential quadratic programming methods in Matlab (v 7.9.0; The Mathworks, Natick, MA). Student's t -tests were used to examine differences between sarcomere length for most variables using Matlab and SPSS (IBM Statistics, Chicago, IL). MgATP-dependent parameter estimates from modulus data fit to Eq. 1 were analyzed using linear mixed models with MgATP as the repeated measure, followed by a least significant difference post hoc comparison of the means using SPSS. Statistical significance is reported at $P < 0.05$.

RESULTS

Myocardial strips show greater passive tension and increased Ca²⁺-sensitivity of developed tension at longer sarcomere length. Across the entire pCa range (8.0–4.8), absolute tension values were greater at 2.2 µm sarcomere length vs. 1.9 µm sarcomere length (Fig. 1A and Table 1). These increased absolute tension values were largely driven by the greater relaxed tension values at 2.2 µm sarcomere length. Developed tension values were also greater for 2.2 vs. 1.9 µm sarcomere lengths at Ca²⁺ levels greater than pCa 5.6 (Fig. 1B and Table 1). Thus Ca²⁺-sensitivity of the tension-pCa relationship was also greater at 2.2 µm sarcomere length, showing an ≈ 0.1 increase in pCa₅₀ for 2.2 µm sarcomere length compared with 1.9 µm sarcomere length. Although there were no statistical differences in the slopes of the tension-pCa relationships, the high Hill coefficients of 6.47 ± 0.94 and 8.68 ± 1.45 for 1.9

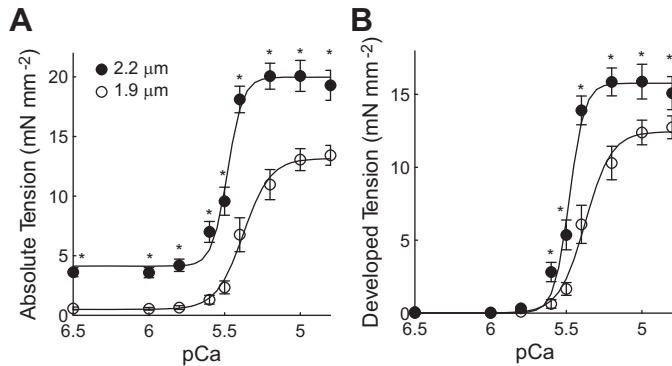


Fig. 1. Absolute tension-pCa (A) and Ca^{2+} -activated, developed tension-pCa (B) relationships at 1.9 and 2.2 μm sarcomere lengths. Lines represent Hill fits to the tension data. $*P < 0.05$ between the 2 sarcomere lengths.

and 2.2 μm sarcomere lengths indicate a highly cooperative response once Ca^{2+} -activated tension started to develop.

Myocardial strips demonstrate slowed cross-bridge cycling at longer sarcomere length. Under relaxed conditions (pCa 8.0), viscoelastic modulus values were greater in myocardial strips that were stretched to 2.2 μm sarcomere length vs. 1.9 μm sarcomere length (Fig. 2). Specifically, elastic moduli values were greater for 2.2 μm sarcomere length fibers across the entire frequency range, and viscous moduli values were greater for 2.2 μm sarcomere length at frequencies greater than 8.5 Hz. Under rigor conditions (pCa 4.8 and >0.001 mM MgATP), viscous and elastic moduli values did not differ at any individual frequency between the two sarcomere lengths. Moreover, there was a large increase in viscoelastic mechanical stiffness of the myocardium at rigor versus relaxed conditions, which illustrates the cross-bridge contribution to myofilament viscoelasticity from strong cross-bridge binding. These relaxed and rigor data combine to illustrate three characteristics of the skinned myocardial strips. First, the increased viscoelastic mechanical stiffness at longer sarcomere length that occurs under relaxed conditions suggests that this difference stems from passive elements of the sarcomere being stretched or extended more at 2.2 vs. 1.9 μm sarcomere length (i.e., titin and collagen) (21). Second, similar moduli values under rigor conditions indicate similar levels of cross-bridge binding during rigor for both sarcomere lengths. Finally, there were no measurable differences in cross-bridge stiffness between the two sarcomere lengths (calculated as the difference between rigor and relaxed moduli values; data not shown),

Table 1. Characteristics of tension-pCa relationships in rat myocardium at 1.9 and 2.2 μm sarcomere lengths

	1.9 μm	2.2 μm
$T_{\min}$, mN mm^{-2}	0.67 ± 0.14	$4.20 \pm 0.50^*$
$T_{\max}$, mN mm^{-2}	13.42 ± 0.82	$19.29 \pm 1.26^*$
T_{dev} , mN mm^{-2}	12.74 ± 0.78	$15.55 \pm 1.13^*$
pCa ₅₀	5.36 ± 0.04	$5.45 \pm 0.01^*$
n_H	6.47 ± 0.94	8.68 ± 1.45
Max_{fit} , mN mm^{-2}	12.56 ± 0.80	$15.54 \pm 1.13^*$

Values are means $\pm$ SE. $T_{\min}$, absolute tension value at pCa 8.0; $T_{\max}$, absolute tension value at pCa 4.8; T_{dev} , Ca^{2+} -activated, developed tension ($T_{\max} - T_{\min}$). Max_{fit} , pCa₅₀, and n_H represent fit parameters to a 3-parameter Hill equation for the T_{dev} -pCa relationship: $T_{\text{dev}}(\text{pCa}) = \text{Max}_{\text{fit}}/[1 + 10^{n_H(\text{pCa} - \text{pCa}_{50})}]$. $*P < 0.05$ between 1.9 and 2.2 μm sarcomere lengths.

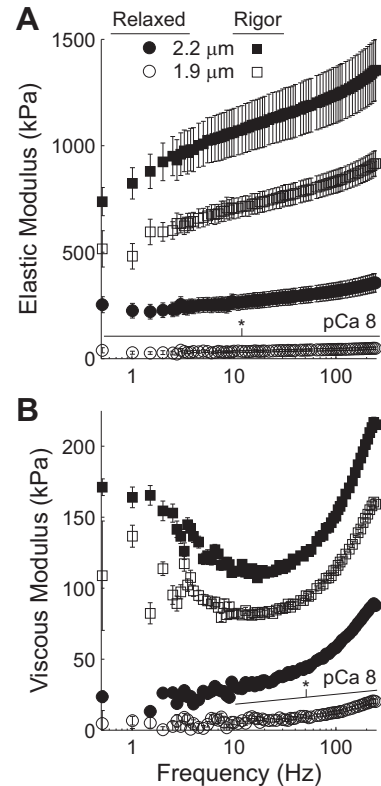


Fig. 2. Elastic (A) and viscous (B) moduli are plotted against frequency for relaxed (pCa 8.0) and rigor (pCa 4.8, MgATP < 0.001 mM) conditions at 1.9 and 2.2 μm sarcomere lengths. Moduli values were different as frequency varied under relaxed conditions, but not under rigor conditions ($*P < 0.05$ between the 2 sarcomere lengths).

demonstrating that bound cross-bridges exhibit similar viscoelastic mechanical characteristics under rigor conditions at both sarcomere lengths.

At maximal calcium activation (pCa 4.8) and 5 mM MgATP, elastic moduli values were greater for myocardial strips stretched to 2.2 μm sarcomere length at frequencies greater than 12 Hz (Fig. 3A). Viscous moduli values were greater between 4 and 13 Hz for myocardial strips stretched to 2.2 vs. 1.9 μm sarcomere length (Fig. 3B). The increased magnitude of the moduli values at 2.2 μm sarcomere length demonstrates increased myocardial viscoelasticity that longer sarcomere length that arises from greater cross-bridge binding at 2.2 vs. 1.9 μm sarcomere length, greater stretch or distortion of passive elements of the sarcomere as force develops at 2.2 vs. 1.9 μm sarcomere length (i.e., myofilaments, titin, and collagen), and prolonged cross-bridge attachment at 2.2 vs. 1.9 μm sarcomere length that effectively appears as increased cross-bridge binding to augment moduli values. Although all these mechanisms contribute to elevated myocardial viscoelasticity at 2.2 μm sarcomere length, our measurements suggest that greater cross-bridge binding at 2.2 vs. 1.9 μm sarcomere length is the primary source for increased myocardial viscoelasticity at 2.2 μm sarcomere length, since the magnitude-difference in elastic and viscous moduli values between the two sarcomere lengths is larger at maximally activated conditions (Fig. 3, A and B) versus relaxed conditions (Fig. 2, A and B). This interpretation is also supported by the elevated devel-

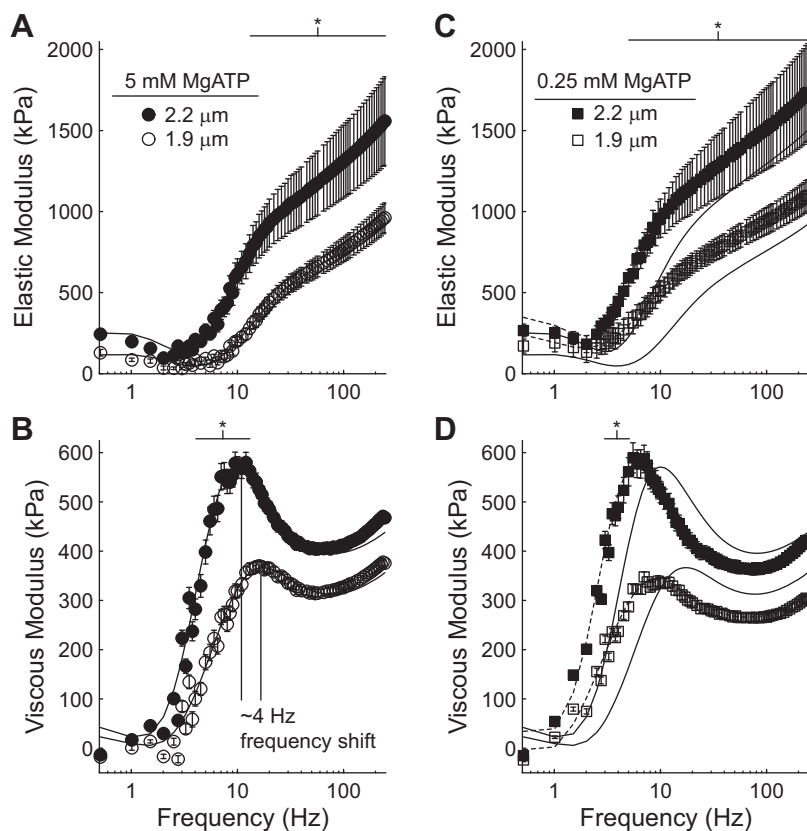


Fig. 3. Elastic (A) and viscous (B) moduli are plotted against frequency for maximally activated conditions (pCa 4.8 and 5 mM MgATP) at 1.9 and 2.2 μm sarcomere lengths. As [MgATP] was reduced to 0.25 mM, elastic (C) and viscous (D) moduli vs. frequency relationships are shifted toward lower frequencies for both sarcomere lengths. Solid lines represent fits to Eq. 1 for each sarcomere length at 5 mM MgATP, and dashed lines represent fits to Eq. 1 for each sarcomere length at 0.25 mM MgATP. Moduli values differed between the 2 sarcomere lengths as frequency varied ($*P < 0.05$), and the vertical lines (B) illustrate the maximal viscous moduli value at each sarcomere length.

oped tension measurements at longer sarcomere length under maximally activated conditions (Fig. 1B).

In addition, the moduli-frequency relationships are shifted toward lower frequencies for the 2.2 μm sarcomere length measurements at maximally activated conditions (Fig. 3, A and B). As a whole, the characteristic peaks and dips in these moduli-frequency relationships indicate a broad metric of cross-bridge cycling kinetics, where a shift toward lower frequencies in these relationships demonstrates slower cross-bridge cycling and a shift toward higher frequencies demonstrates faster cross-bridge cycling. An example of this frequency shift is highlighted by the vertical lines in Fig. 3B for the viscous moduli versus frequency relationships, which show a peak value at 12 Hz for 2.2 μm sarcomere length and a peak value at 16 Hz for 1.9 μm sarcomere length. This shift toward lower frequencies in the moduli-frequency response indicates slower cross-bridge cycling for the myocardial strips at 2.2 μm sarcomere length compared with the strips at 1.9 μm sarcomere length.

Consistently, there was a shift toward lower frequencies in the moduli-frequency relationships for both sarcomere lengths as [MgATP] decreased from 5 to 0.25 mM at pCa 4.8 (Fig. 3, C and D). This frequency shift demonstrates slowed cross-bridge cycling at 0.25 mM vs. 5 mM MgATP, as would be expected due to the prolonged time that myosin occupies the rigor state before binding MgATP at 0.25 mM MgATP. Under these conditions, myocardial strips stretched to 2.2 μm sarcomere length had greater elastic moduli values at frequencies greater than 5 Hz (Fig. 3C) and greater viscous moduli values between 3 and 5 Hz (Fig. 3D). Again these elevated moduli values stem from greater cross-bridge binding at 2.2 vs. 1.9 μm

sarcomere lengths, even at 0.25 mM MgATP. The moduli-frequency relationships at 0.25 mM MgATP were also shifted toward lower frequencies at 2.2 μm vs. 1.9 μm sarcomere lengths, with the peak viscous moduli value occurring at 5.5 Hz at 2.2 μm sarcomere length and at 9.5 Hz at 1.9 μm sarcomere length (Fig. 3D). Thus, the frequency shift in the moduli-frequency relationships continues to demonstrate slower cross-bridge cycling for the myocardial strips at 2.2 μm sarcomere length compared with the strips at 1.9 μm sarcomere length, even at a relatively low, subphysiological [MgATP].

Myosin cross-bridges release MgADP more slowly at longer sarcomere length. Maximally activated (pCa 4.8) moduli values were fit to Eq. 1 to extract model parameters related to myocardial viscoelasticity, cross-bridge binding, and cross-bridge kinetics from 0.05 to 5 mM MgATP. Model parameters are plotted against [MgATP] in Fig. 4, with P values listed above each panel for significant effects of [MgATP], sarcomere length, or the [MgATP] $\times$ sarcomere length interaction. As [MgATP] decreased, values for A, B, and C increased and the magnitude of these parameters was consistently greater at 2.2 vs. 1.9 μm sarcomere length, showing significant effects of [MgATP] and sarcomere length (Fig. 4, A, C, and E). There was also an interaction between [MgATP] $\times$ sarcomere length for A, showing a greater increase in the magnitude of A as [MgATP] decreased at 2.2 μm sarcomere length (Fig. 4A). These MgATP-dependent increases correspond to a greater number of bound cross-bridges being attached longer at 2.2 vs. 1.9 μm sarcomere length, which enhances viscoelastic myocardial stiffness as [MgATP] decreased. As introduced above, the length-dependent increases in B and C indicate greater cross-bridge binding at 2.2 vs. 1.9 μm sarcomere length, rather

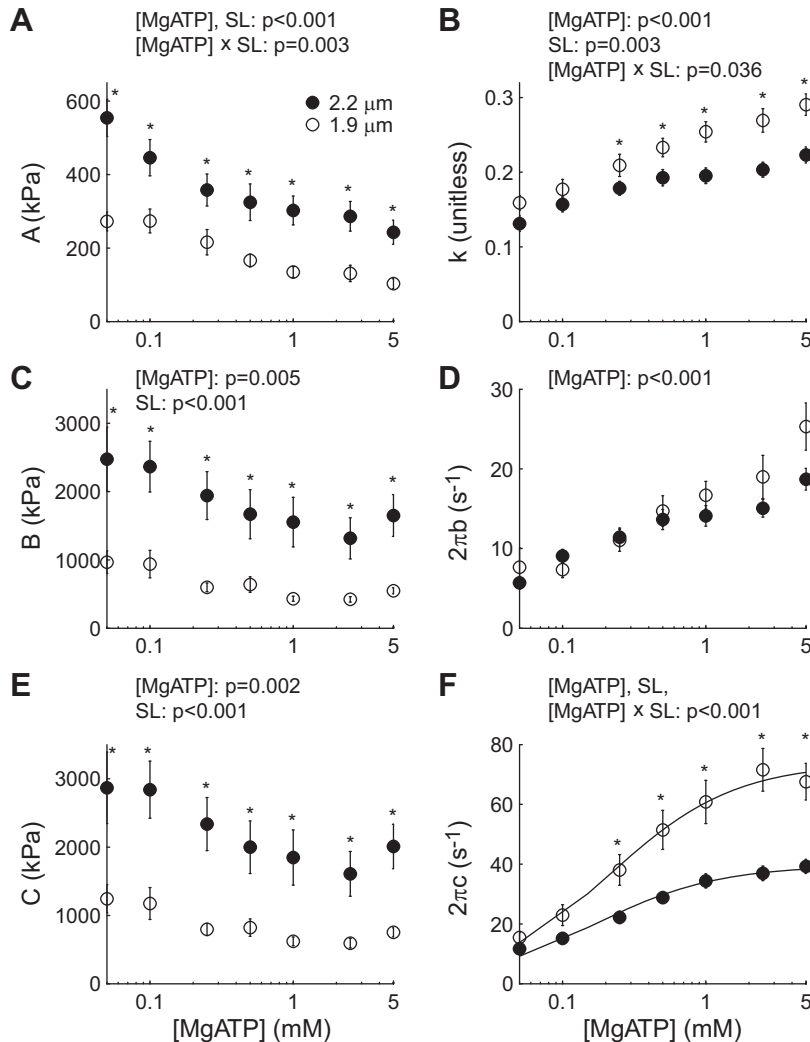


Fig. 4. Effects of [MgATP] on parameter fits to Eq. 1 for the complex moduli at maximal Ca^{2+} -activation (pCa 4.8) for 1.9 and 2.2 μm sarcomere lengths (SL). Myocardial viscoelastic stiffness rose and became increasingly elastic as [MgATP] decreased, as reflected by the increase in the parameter A (A) and decrease in the parameter k (B) for both sarcomere lengths. The rate of force redevelopment (D), $2\pi b$, and the rate of cross-bridge detachment (F), $2\pi c$, decreased as [MgATP] decreased for both sarcomere lengths. Magnitude parameters for the B-process (C) and the C-process (E) also increased as [MgATP] decreased, which indicates the expected increase in cross-bridge binding as [MgATP] was titrated toward rigor. P values from mixed models analysis of each parameter-[MgATP] relationship are listed above each panel for main effects of [MgATP], sarcomere length, or a [MgATP] $\times$ SL interaction, where significant ($*P < 0.05$ between 1.9 and 2.2 μm sarcomere lengths.).

than any differences in cross-bridge stiffness, because elastic and viscous moduli values did not differ under rigor conditions.

As [MgATP] decreased, k values also decreased and the magnitude of k was less for 2.2 vs. 1.9 μm sarcomere length, with significant effects of MgATP and sarcomere length (Fig. 4B). Smaller k values indicate an increased elastic characteristic of the myocardium as [MgATP] decreased, and for 2.2 vs. 1.9 μm sarcomere length. A portion of this MgATP-dependent reduction in k corresponds to an increased number of strongly bound cross-bridges as [MgATP] decreased. Likewise the [MgATP] $\times$ sarcomere length interaction suggests that the length-dependent increases in cross-bridge binding that enhance myocardial elasticity at 2.2 μm sarcomere length are greatest at 5 mM MgATP and diminish as [MgATP] was titrated toward rigor conditions. However, a portion of this response also stems from slower cross-bridge cycling at 2.2 μm , which could enhance the apparent elasticity of the myocardium at 2.2 vs. 1.9 μm sarcomere length and would contribute most greatly to this response at high [MgATP] where kinetics differ most greatly between the two sarcomere length (Fig. 4F).

As [MgATP] decreased the cross-bridge attachment rate ($2\pi b$; Fig. 3D) and detachment rate ($2\pi c$; Fig. 4F) also

decreased. There was not a significant effect of sarcomere length on the rate of cross-bridge attachment. In contrast, the rate of cross-bridge detachment was slower at 2.2 vs. 1.9 μm sarcomere length. Cross-bridge detachment rate was also more sensitive to variations in MgATP at 1.9 μm sarcomere length than 2.2 μm sarcomere length ($P < 0.001$ for the [MgATP] $\times$ sarcomere length interaction), but this finding may follow from faster detachment rates near saturating [MgATP] that facilitate a greater dynamic range for cross-bridge detachment rate versus [MgATP] at 1.9 μm sarcomere length (Fig. 4F).

Fitting the $2\pi c$ -MgATP relationship to Eq. 2 links cross-bridge detachment rate with fundamental steps of the cross-bridge cycle, specifically the cross-bridge rate of MgADP release (k_{ADP}), and the cross-bridge rate of MgATP binding (k_{ATP}) (61, 64). These fits (solid lines, Fig. 4F) demonstrate that the slowed cross-bridge detachment rate at 2.2 μm sarcomere length stems from a slower MgADP release rate, since MgATP binding rates were not statistically different between both sarcomere lengths (Table 2). These findings indicate that increasing sarcomere length from 1.9 to 2.2 μm slowed MgADP release rate by 45% in myocardial strips from rat papillary muscles. This slowed MgADP release rate increased cross-bridge attachment time ($t_{\text{on}} = 1/2\pi c$) by 70% at 2.2 μm compared with 1.9 μm sarcomere length (Fig. 5).

Table 2. Estimates of myosin cross-bridge kinetics from fits of the cross-bridge detachment rate ($2\pi\tau$) versus MgATP relationships to Eq. 2 for 1.9 and 2.2 μm sarcomere lengths

	1.9 μm	2.2 μm
k_{ADP} , s^{-1}	$74.21 \pm 6.99^*$	$40.74 \pm 2.69^*$
k_{ATP} , mM/s	361.28 ± 63.13	297.71 ± 59.09
$[\text{MgATP}]_{50}$, μM	282.62 ± 54.72	218.03 ± 42.39
t_{ADP} , ms	$13.47 \pm 1.27^*$	$24.55 \pm 1.62^*$

Values are means $\pm$ SE. k_{ADP} , cross-bridge MgADP release rate; k_{ATP} , cross-bridge MgATP binding rate; $[\text{MgATP}]_{50} = k_{\text{ADP}}/k_{\text{ATP}}$, MgATP concentration at half-maximal detachment rate; $t_{\text{ADP}} = 1/k_{\text{ADP}}$, myosin cross-bridge attachment duration for the MgADP state. $^*P < 0.05$ between 1.9 and 2.2 μm sarcomere lengths.

Length-dependent effects on cross-bridge kinetics decreased at submaximal pCa levels. During physiological contraction, intracellular $[\text{Ca}^{2+}]$ and sarcomere length are dynamic variables that change throughout a heartbeat. To extrapolate our findings from maximally activated Ca^{2+} conditions toward lower Ca^{2+} conditions that exist in the heart during physiological contraction, we compared stochastic length-perturbation measurements near the pCa_{50} value for each sarcomere length where developed tension levels were similar (Fig. 6). At these matched tension levels, elastic and viscous moduli values were not different between the two sarcomere lengths, but the characteristic peaks and dips in these moduli-frequency relationships were shifted toward lower frequencies for 2.2 vs. 1.9 μm sarcomere length. The peak viscous moduli values occurred at 13 Hz for 2.2 μm sarcomere length and at 16 Hz for 1.9 μm sarcomere length, which broadly demonstrates slower cross-bridge cycling at 2.2 vs. 1.9 μm sarcomere length that is consistent with observations under maximally activated conditions (Fig. 3, A and B).

When compared with the results at pCa 4.8, these moduli values near pCa_{50} have a lower magnitude and higher variance. These differences stem from less cross-bridge binding at submaximal pCa levels, which reduces tension and the signal-to-noise ratio in our stochastic length-perturbation measurements. The frequency-dependence of the elastic and viscous moduli is similar between pCa 5.4 and pCa 4.8 at 1.9 μm sarcomere length, which suggests that myosin kinetics were similar at submaximal and maximally activated conditions for this shorter sarcomere length. However, there was a small shift toward higher frequencies in the moduli-frequency response at pCa 5.5 vs. pCa 4.8 at 2.2 μm sarcomere length, which

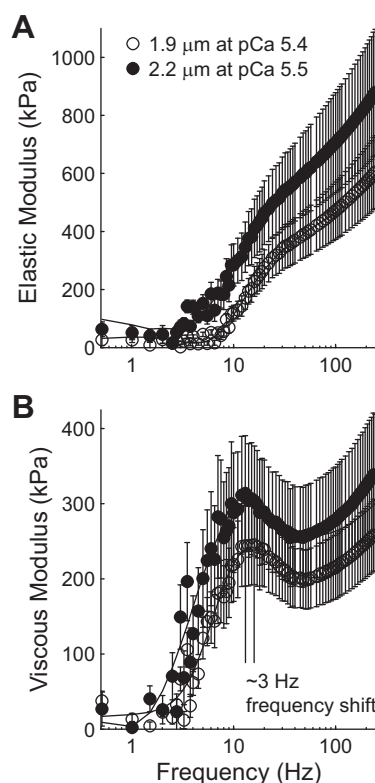


Fig. 6. Elastic (A) and viscous (B) moduli are plotted against frequency for submaximally activated conditions that were near the pCa_{50} value for each sarcomere length (i.e., pCa 5.4 for 1.9 μm sarcomere length and pCa 5.5 for 2.2 μm sarcomere length). Solid lines represent fits to Eq. 1 for each sarcomere length, with the vertical lines (B) illustrating the maximal viscous moduli value at each sarcomere length.

suggests a slight increase in cross-bridge cycling kinetics at submaximal pCa levels for this longer sarcomere length.

These moduli values at submaximal pCa (Fig. 6) were fit to Eq. 1 to extract model parameters related to myocardial viscoelasticity, cross-bridge binding, and cross-bridge kinetics (Table 3). Consistent with the findings at pCa 4.8, A was greater and k was smaller at 2.2 vs. 1.9 μm sarcomere length at these submaximal pCa levels. Again this demonstrates the viscoelastic myocardial stiffness was greater, and showed a more elastic response, at 2.2 μm sarcomere length, which is consistent with greater extension or stretch of the passive elements of the sarcomere at longer sarcomere length. Overall, B and C were

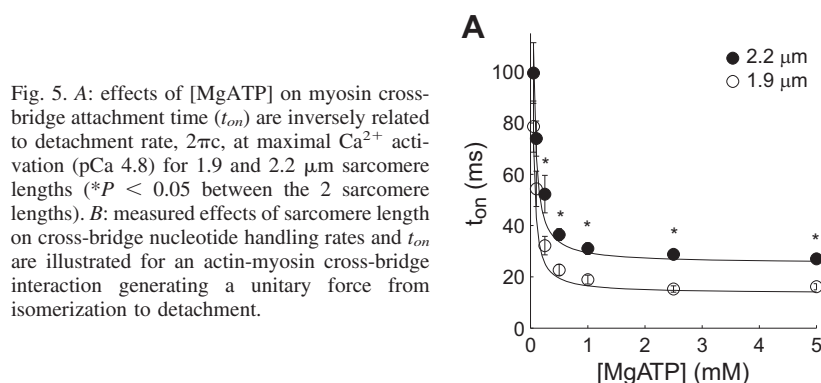


Fig. 5. A: effects of $[\text{MgATP}]$ on myosin cross-bridge attachment time (t_{on}) are inversely related to detachment rate, $2\pi\tau$, at maximal Ca^{2+} activation (pCa 4.8) for 1.9 and 2.2 μm sarcomere lengths ($^*P < 0.05$ between the 2 sarcomere lengths). B: measured effects of sarcomere length on cross-bridge nucleotide handling rates and t_{on} are illustrated for an actin-myosin cross-bridge interaction generating a unitary force from isomerization to detachment.

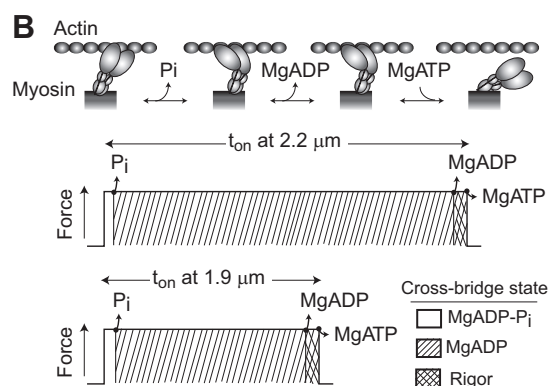


Table 3. Model parameters from fits to Eq. 1 for elastic and viscous moduli values at submaximal pCa levels shown in Fig. 6 for 1.9 and 2.2 μm sarcomere lengths

	1.9 μm	2.2 μm
A, kPa	40.31 ± 9.36	$113.59 \pm 25.91^*$
K, unitless	0.35 ± 0.02	$0.26 \pm 0.02^*$
B, kPa	316.12 ± 28.36	$224.89 \pm 21.92^*$
$2\pi b$, s^{-1}	28.99 ± 2.89	$14.00 \pm 1.94^*$
C, kPa	411.53 ± 30.86	$285.15 \pm 25.74^*$
$2\pi c$ (s^{-1})	64.12 ± 2.72	$55.57 \pm 2.13^*$

Values are means $\pm$ SE. * $P < 0.05$ between 1.9 and 2.2 μm sarcomere lengths.

smaller for both sarcomere lengths at these submaximal pCa levels, compared with the values at maximally activated conditions. The reduced magnitude of these parameter values follows from reduced cross-bridge binding at submaximal pCa levels, as would be expected. Interestingly, the values of both B and C were greater for 1.9 μm sarcomere length at pCa 5.4 vs. 2.2 μm sarcomere length at pCa 5.5, which indicates there was greater cross-bridge binding for 1.9 μm sarcomere length at pCa 5.4. This slight increase in cross-bridge binding for 1.9 μm sarcomere length at pCa 5.4 is consistent with a slightly elevated developed tension under these conditions, in comparison with the developed tension at 2.2 μm sarcomere length at pCa 5.5 (Fig. 1B). Rates of cross-bridge attachment ($2\pi b$) and detachment ($2\pi c$) were similar at 1.9 μm sarcomere length for pCa 5.4 (Table 3) and pCa 4.8 (Fig. 4, D and F), suggesting that cross-bridge cycling kinetics were not affected by decreases in $[\text{Ca}^{2+}]$ at 1.9 μm sarcomere length. In contrast, cross-bridge attachment rate decreased and cross-bridge detachment rate increased for 2.2 μm sarcomere length at pCa 5.5 vs. pCa 4.8 (Fig. 4, D and F), suggesting that reductions in thin-filament activation at submaximal pCa levels influence cross-bridge kinetics at 2.2 μm sarcomere length. Therefore, t_{on} was roughly 15% longer at 2.2 vs. 1.9 μm sarcomere length at these submaximal pCa values (18.26 ± 0.71 ms vs. 15.97 ± 0.74 ms; $P = 0.042$) versus the $\sim 70\%$ increase for 2.2 vs. 1.9 μm sarcomere length at pCa 4.8 (27.02 ± 1.84 vs. 16.12 ± 1.18 ; $P < 0.001$) that is illustrated in Fig. 5. Altogether these findings show that sarcomere length-dependent effects on cross-bridge kinetics persist at submaximal pCa levels, although the kinetic differences are not as great at submaximal pCa conditions compared with maximally activated conditions.

DISCUSSION

We find that myosin cross-bridge cycling slows as sarcomere length increases, and this slowing is predominantly driven by slower cross-bridge MgADP release rate at 2.2 vs. 1.9 μm sarcomere length in skinned papillary muscle strips from rats. These data also show that myosin attachment time (t_{on}) increases at longer sarcomere length, which may amplify cross-bridge contributions to thin-filament activation and augment force production at longer sarcomere lengths. Consistently, shorter t_{on} values may promote thin-filament deactivation and enhance relaxation dynamics at shorter sarcomere lengths. Length-dependent effects on cross-bridge kinetics were greater at maximal (pCa 4.8) versus submaximal (pCa 5.5–5.4) activation levels, with submaximal activation having a minimal influence on cross-bridge kinetics at 1.9 μm sarco-

mere length and some influence on cross-bridge cycling kinetics at 2.2 μm sarcomere length. Thus thin-filament activation may affect cross-bridge attachment and detachment rates differently since sarcomere length varies throughout a heartbeat. Altogether, this study further demonstrates that length-dependent cross-bridge kinetics may contribute, in part, to the Frank-Starling relationship in the heart.

Cross-bridge contributions to thin-filament activation. As sarcomere length increased from 1.9 to 2.2 μm at maximally activated conditions (pCa 4.8 and 5 mM MgATP), t_{on} increased $\approx 70\%$, from 16 to 27 ms at 17°C . At submaximal activation levels near the pCa₅₀ value of the tension-pCa relationship for 1.9 and 2.2 μm sarcomere length, t_{on} increased $\approx 15\%$, from 16 to 18 ms, due to an increased cross-bridge detachment rate at pCa 5.5 vs. pCa 4.8 for the longer sarcomere length. These findings support the idea that strongly bound cross-bridges could augment or amplify force production at submaximal pCa levels by stabilizing thin-filament activation in the open state (45). Our force-pCa relationships (Fig. 1) and myosin kinetics data (Figs. 4–6 and Table 3) suggest that the cooperative contribution to thin-filament activation from strongly bound cross-bridges increases with sarcomere length. Previous studies have linked greater Ca^{2+} -affinity of troponin C and/or greater opening of the NH_2 -terminus of troponin C with increases in strong cross-bridge binding (27, 41, 56, 65), and our findings imply this cooperative activation pathway may become stronger at longer sarcomere lengths. Thus sarcomere length-dependent slowing of cross-bridge detachment could prolong t_{on} , thereby representing an intrinsic feedback mechanism at the myosin motor per se that could augment force production and thin-filament activation at longer sarcomere lengths.

Tension development rates versus nucleotide handling rates. The majority of studies investigating sarcomere length-dependent cross-bridge kinetics have used relatively large amplitude release-restretch protocols to assess the cross-bridge rate of tension redevelopment (k_{tr}), in comparison with the low amplitude strains used for stochastic length perturbation analysis ($<0.15\%$ muscle length). Slower k_{tr} values have been observed at longer sarcomere length (1, 24, 39, 47, 53, 57), whereas others studies saw no change in k_{tr} with sarcomere length (11, 23). Stelzer and Moss (57) also used a 1% step-stretch protocol to measure a $\sim 50\%$ slowing of the cross-bridge detachment rate and no change in attachment rate as sarcomere length increased from 1.9 to 2.25 μm in maximally activated mouse myocardial strips. We found no effect of sarcomere length on the rate of cross-bridge attachment ($2\pi b$) as [MgATP] varied, but slowed cross-bridge detachment ($2\pi c$) as sarcomere length increased (Fig. 4), which agrees well with the step-stretch kinetics measured by Stelzer and Moss (57). Because k_{tr} is thought to be a summation of the cross-bridge attachment and cross-bridge detachment rates (5), our data support the idea that k_{tr} decreases with sarcomere length due to slowed cross-bridge detachment rate with an insignificant or minimal change in cross-bridge attachment rate (1, 57).

By measuring cross-bridge kinetics as [MgATP] varied, we were able to assess whether cross-bridge rates of MgADP release or MgATP binding are influenced by sarcomere length. Under saturating [MgATP] t_{on} is dominated by the MgADP release rate, as the rate limiting step of the cross-bridge cycle. However, as [MgATP] decreases cross-bridge detachment rate

becomes increasingly dominated by the second-order MgATP binding process, thereby allowing an estimate of cross-bridge nucleotide handling rates (i.e., MgADP release and MgATP binding). There was not a significant difference between cross-bridge MgATP binding rates at 1.9 vs. 2.2 μm sarcomere lengths, and the apparent cross-bridge affinity for MgATP was similar at both sarcomere lengths (as evidenced by similar $[\text{MgATP}]_{50}$ values in Table 2). However, there was a significant $\text{SL} \times \text{MgATP}$ interaction term for cross-bridge detachment rate ($P < 0.001$ for $2\pi c$; Fig. 4F), which likely stems from the greater overall slowing of cross-bridge detachment rate as $[\text{MgATP}]$ decreased at 1.9 μm sarcomere length. Therefore, we conclude that slowed MgADP release rate is the predominate step of the cross-bridge cycle that is responsible for the length-dependent increase in t_{on} (Fig. 5B).

Cross-bridge detachment rate and t_{on} values become saturated at $[\text{MgATP}]$ above 2.5 mM (Fig. 4F and Fig. 5), and we expect that cross-bridges would be operating near saturating MgATP levels in a working heart. At these $[\text{MgATP}]$, the duration of cross-bridge binding events becomes dominated by the cross-bridge detachment rate, which is closely linked to the cross-bridge rate of MgADP release. Because the rate that cross-bridges bind MgATP is a second order process, it will be affected by $[\text{MgATP}]$, but $[\text{MgATP}]$ has to become quite low to affect t_{on} as much as MgADP release rates. Even at 2.5 mM MgATP, cross-bridge rates of MgATP binding are an order of magnitude faster (≈ 900 and 750 s^{-1} for 1.9 and 2.2 μm sarcomere lengths) than MgADP release rates (Table 2). One can estimate the $[\text{MgATP}]$ that is required for cross-bridges to spend equal time in the MgADP state (t_{ADP} ; Table 2) and in the rigor state waiting for MgATP to bind (t_{+ATP}): $t_{+ATP} = 1/(k_{+ATP} \times [\text{MgATP}])$, which would occur at ≈ 206 and $137 \mu\text{M}$ MgATP for 1.9 and 2.2 μm sarcomere lengths at maximally activated conditions. Although the duration of time that a bound cross-bridge spends in the MgADP-Pi, MgADP, and rigor state can be estimated from our measurements, as illustrated in Fig. 5, the cross-bridge spends most of its time waiting for MgADP to dissociate at $[\text{MgATP}] \geq 2.5 \text{ mM}$. Therefore, it is primarily the rate of MgADP release that dominates the kinetics of a cross-bridge binding event at saturating MgATP levels in cardiac muscle strips.

The length-dependent cross-bridge kinetics that we acquired at 17°C can be extrapolated toward the physiological temperature of 37°C by accounting for the approximate sixfold change in the frequency-dependence of the moduli-frequency relationships that occurs with a 20°C increase in temperature in skinned murine myocardial strips (50). Therefore, myosin detachment kinetics would increase and estimated t_{on} values would decrease, as $t_{on} = (2\pi c)^{-1}$, but the relative increases in t_{on} as sarcomere length increases from 1.9 to 2.2 μm would remain similar. Estimated t_{on} values at physiological temperature would be roughly 2.5 and 4.2 ms for 1.9 to 2.2 μm sarcomere lengths at maximally activated conditions ($=\text{pCa}$ 4.8 and 5 mM MgATP; Fig. 4F), maintaining the 70% increase in cross-bridge attachment time at the longer sarcomere length. Similarly, estimates of t_{on} at submaximal pCa levels (Fig. 6 and Table 3) would still increase roughly 15% from 2.6 to 3 ms as sarcomere length increases from 1.9 to 2.2 μm at 37°C .

Similar measurements have been made by Wang et al. (64) using sinusoidal length-perturbation analysis in skinned papillary muscle strips from Wistar-Kyoto rats, where they mea-

sured cross-bridge rates of MgADP release and MgATP binding of $65 \pm 7.9 \text{ s}^{-1}$ and $108 \pm 10 \text{ mM/s}$ under maximally activated conditions at 2.2 μm sarcomere length and 17°C . We find faster rates of MgADP release and MgATP binding at 2.2 μm sarcomere length (Table 2) than with these prior measurements, although the order of magnitude was similar among each respective set of nucleotide handling rates in both studies. We have illustrated some systematic differences between sinusoidal- and stochastic-length perturbation analysis, although the differences are relatively small when oscillatory amplitude distribution, frequency range, sampling frequency, solution conditions, and experimental equipment are matched among single muscle fibers (60).

Myofilament viscoelasticity as sarcomere length increased. The viscoelastic stiffness characteristics of the myocardium that arise from passive elements of myofilament lattice are augmented at 2.2 vs. 1.9 μm sarcomere length. This arises in the measured tension values (Fig. 1A) as well as elastic and viscous moduli values (Fig. 2) under relaxed conditions. These responses are primarily attributed to extension of titin and collagen as myocardial strips were stretched to 2.2 μm (21). However, under rigor conditions there were not significant differences in the viscoelasticity of the myocardium between the two sarcomere lengths, which also suggests no differences in cross-bridge stiffness between the two sarcomere lengths. The viscoelastic differences between relaxed and rigor conditions indicate considerable cross-bridge contributions to the passive mechanical characteristics of the myocardium in rigor, showing that the cross-bridge component dominates the titin and collagen contributions to myocardial viscoelasticity in rigor.

Because the fiber was maximally activated to pCa 4.8, the contributions to myocardial viscoelasticity are composed of passive (A-process) and dynamic elements (B- and C-processes), which arise from passive elements (i.e., titin, collagen, myofilaments) and enzymatic elements (i.e., cycling cross-bridges) within the myocardium (Fig. 4). Similar to the measurements under relaxed conditions, this increased A value under activated conditions is primarily due to greater titin and collagen extension/stretch at longer sarcomere length. The magnitudes of B and C were also greater at 2.2 vs. 1.9 μm sarcomere length, indicating greater cross-bridge binding at longer sarcomere length. A portion of this increased cross-bridge contribution to myocardial viscoelasticity could stem from kinetic differences between 2.2 vs. 1.9 μm sarcomere length, such that longer t_{on} values at longer sarcomere length effectively show up as greater cross-bridge binding. Altogether, these data demonstrate increased viscoelastic stiffness characteristics for maximally activated myocardial strips at 2.2 vs. 1.9 μm sarcomere length due to greater titin and collagen extension and greater cross-bridge binding at 2.2 μm sarcomere length.

Physical factors affecting cross-bridge kinetics throughout a heartbeat. The finding that slower cross-bridge MgADP release rate prolongs t_{on} as sarcomere length increases is consistent with our previous observations that cross-bridge cycling slows as thick-to-thin-filament spacing decreased in insect flight muscle (59). Cross-bridge cycling also slowed as thick-to-thin filament spacing decreased in vertebrate and invertebrate muscle fibers that were osmotically compressed with Dextran (33, 40, 56). Similarly, increases in sarcomere length

have been shown to decrease thick-to-thin filament spacing and slow cross-bridge kinetics in skinned (1) and intact (47) cardiac muscle preparations. However, Konhilas et al. (37) demonstrated that similar decreases in myofilament lattice spacing from increased osmotic compression versus increased sarcomere length did not produce similar increases in Ca^{2+} -sensitivity of the tension-pCa relationship, although they did not report any measure of cross-bridge kinetics. Although it is clear that thick-to-thin filament spacing values are known to differ between skinned muscle fibers and intact or in vivo muscle preparations (31, 37, 43, 56), myofilament lattice spacing consistently decreases with increased sarcomere length among all of these preparations. Therefore, increases in sarcomere length will accompany decreases in thick-to-thin filament spacing to some degree, both of which represent structural changes to the myofilament lattice architecture that could enhance the probability of cross-bridge binding and contribute to slower cross-bridge detachment at longer sarcomere length.

Reductions in lattice spacing that concomitantly occur with increased sarcomere length could also alter the volumetric constraints experienced by a myosin head throughout the cross-bridge cycle. These constraints could influence strain-dependent cross-bridge cycling kinetics (as further discussed below) as well as the micro-environment solution conditions in which a myosin cross-bridge operates. The $\sim 5\%$ decrease in lattice spacing at 0% dextran that was reported by Konhilas et al. (37) (44 vs. 42 nm between 1.9 and 2.2 μm sarcomere lengths) would produce a $\sim 10\%$ reduction in solution volume surrounding a thick-filament and the associated six thin-filaments with which it may bind. Thus it is possible that slowed MgADP release at longer sarcomere length could be affected by MgADP accumulation in the micro-environment directly surrounding a thick-filament due to a slight reduction in solution volume as lattice spacing decreases. However, we attempt to mitigate reduced intrafibrillar [MgATP] or increased intrafibrillar [MgADP] during an experiment via relatively high concentrations of phosphocreatine (30 mM) and creatine kinase (300 U/ml) that should adequately maintain saturating MgATPase rates in a muscle fiber (8). Although we do not directly measure cross-bridge MgATPase, we do not think any aberrant accumulation of MgADP would result from faster MgATPase rates at 2.2 vs. 1.9 μm sarcomere length, since our data supports the idea that myosin duty ratio increases and MgATPase decreases at the longer sarcomere length.

Given that reductions in myofilament lattice spacing between 1.9 and 2.2 μm sarcomere lengths may not solely underlie the length-dependent cross-bridge kinetics that we observe, it is possible that structural changes to the myofilament lattice could also affect strain-dependent or load-dependent cross-bridge transition rates (22, 29, 30, 54, 55). As myofilament lattice spacing dynamically changes throughout a heartbeat, slowed MgADP release rate and prolonged t_{on} could augment contractility at longer sarcomere lengths, as systole begins. As the heart contracts and sarcomere length decreases, faster MgADP release and reduced t_{on} could favor relaxation function and ventricular refilling during diastole. Decreases in lattice spacing at longer sarcomere length could favor the probability of cross-bridge binding and enhance the ability to remain bound longer, which could translate into greater force production at longer sarcomere length (63). Perhaps a slower MgADP release rate enables cross-bridges to bear higher loads

or generate greater forces before detaching from actin at longer sarcomere length (22, 54).

A complementary strain-dependent argument could be constructed for increases in myofilament lattice spacing that occur at shorter sarcomere lengths, which contributes to faster MgADP release rates and shorter t_{on} values at short sarcomere length. Perhaps increases in lattice spacing diminish the probability of cross-bridge binding and enhance strain borne by a cross-bridge while bound to actin, thereby enhancing cross-bridge detachment. This would be consistent with faster MgADP release rates that we observed, if cross-bridges generated lower forces or bore less force at short sarcomere length (22, 54). However, this argument is not entirely consistent with distortion-dependent cross-bridge theory that links greater cross-bridge distortion with greater cross-bridge strain, thereby leading to greater force production at a fixed cross-bridge stiffness value (9, 12, 29, 30, 52, 58). Recent experimental and computational studies showing nonlinear cross-bridge stiffness characteristics and varied contributions to radial versus longitudinal cross-bridge force production also suggest that further studies are required to illustrate a clear relationship between cross-bridge distortion and strain-dependent cross-bridge transition rates (34, 67). Nonetheless, the current study demonstrates that cross-bridge MgADP release varies with sarcomere length, which could enhance systolic contraction as well as diastolic relaxation throughout a heartbeat.

Conclusions

Changes in sarcomere length affect many processes related to thin-filament activation, cross-bridge force production, and contractile deactivation throughout a heartbeat. Our findings show that increasing sarcomere length slows myosin MgADP release rate, prolongs myosin attachment duration, and enhances cross-bridge binding. These changes in cross-bridge kinetics and binding could enhance force generation and stabilize the open or active state of troponin to increase contractility at longer sarcomere lengths. Conversely, faster cross-bridge detachment and diminished cross-bridge binding at short sarcomere lengths could promote relaxation during diastole. Therefore, length-dependent changes in myosin MgADP release rate may be a molecular mechanism that is fundamental to the Frank-Starling law of the heart.

ACKNOWLEDGMENTS

We thank Wenji Dong, Axel Fenwick, and Jim Peters for helpful discussions and comments on previous versions of this manuscript.

GRANTS

These studies were supported by a Beginning Grant in Aide (14BGIA20380385) from the Western States Affiliate of the American Heart Association, as well as New-faculty Seed Grants from Washington State University and the College of Veterinary Medicine.

DISCLOSURES

No conflicts of interest, financial or otherwise, are declared by the author(s).

AUTHOR CONTRIBUTIONS

Author contributions: B.C.T. conception and design of research; B.C.T., J.J.B., and P.O.A. performed experiments; B.C.T., J.J.B., and P.O.A. analyzed data; B.C.T. and J.J.B. interpreted results of experiments; B.C.T. and J.J.B. prepared figures; B.C.T. drafted manuscript; B.C.T. and P.O.A. edited and

revised manuscript; B.C.T., J.J.B., and P.O.A. approved final version of manuscript.

REFERENCES

- Adhikari B, Wang K. Interplay of troponin- and myosin-based pathways of calcium activation in skeletal and cardiac muscle: the use of W7 as an inhibitor of thin filament activation. *Biophys J* 86: 359–370, 2004.
- Adhikari BB, Regnier M, Rivera AJ, Kreutziger KL, Martyn DA. Cardiac length dependence of force and force redevelopment kinetics with altered cross-bridge cycling. *Biophys J* 87: 1784–1794, 2004.
- Allen DG, Kentish JC. The cellular basis of the length-tension relation in cardiac muscle. *J Mol Cell Cardiol* 17: 821–840, 1985.
- Bremel RD, Weber A. Cooperation within actin filament in vertebrate skeletal muscle. *Nat New Biol* 238: 97–101, 1972.
- Brenner B. Effect of Ca^{2+} on cross-bridge turnover kinetics in skinned single rabbit psoas fibers: implications for regulation of muscle contraction. *Proc Natl Acad Sci USA* 85: 3265–3269, 1988.
- Campbell K, Chandra M, Kirkpatrick R, Slinker B, Hunter W. Interpreting cardiac muscle force-length dynamics using a novel functional model. *Am J Physiol Heart Circ Physiol* 286: H1535–H1545, 2004.
- Cazorla O, Wu Y, Irving TC, Granzier H. Titin-based modulation of calcium sensitivity of active tension in mouse skinned cardiac myocytes. *Circ Res* 88: 1028–1035, 2001.
- Cooke R, Bialek W. Contraction of glycerinated muscle fibers as a function of the ATP concentration. *Biophys J* 28: 241–258, 1979.
- Daniel TL, Trimble AC, Chase PB. Compliant realignment of binding sites in muscle: transient behavior and mechanical tuning. *Biophys J* 74: 1611–1621, 1998.
- de Tombe PP, Mateja RD, Tachampa K, Ait Mou Y, Farman GP, Irving TC. Myofilament length dependent activation. *J Mol Cell Cardiol* 48: 851–858, 2010.
- Edes IF, Czuriga D, Csanyi G, Chlopicki S, Recchia FA, Borbely A, Galajda Z, Edes I, van der Velden J, Stienen GJM, Papp Z. Rate of tension redevelopment is not modulated by sarcomere length in permeabilized human, murine, and porcine cardiomyocytes. *Am J Physiol Regul Integr Comp Physiol* 293: R20–R29, 2007.
- Eisenberg E, Hill TL, Chen Y. Cross-bridge model of muscle contraction: quantitative analysis. *Biophys J* 29: 195–227, 1980.
- Fabiato A, Fabiato F. Dependence of the contractile activation of skinned cardiac cells on the sarcomere length. *Nature* 256: 54–56, 1975.
- Farman GP, Allen EJ, Schoenfeldt KQ, Backx PH, de Tombe PP. The role of thin filament cooperativity in cardiac length-dependent calcium activation. *Biophys J* 99: 2978–2986, 2010.
- Farman GP, Walker JS, de Tombe PP, Irving TC. Impact of osmotic compression on sarcomere structure and myofilament calcium sensitivity of isolated rat myocardium. *Am J Physiol Heart Circ Physiol* 291: H1847–H1855, 2006.
- Fitzsimons DP, Moss RL. Strong binding of myosin modulates length-dependent Ca^{2+} activation of rat ventricular myocytes. *Circ Res* 83: 602–607, 1998.
- Ford SJ, Chandra M. Length-dependent effects on cardiac contractile dynamics are different in cardiac muscle containing α - or β -myosin heavy chain. *Arch Biochem Biophys* 535: 3–13, 2013.
- Fuchs F, Martyn DA. Length-dependent Ca^{2+} activation in cardiac muscle: some remaining questions. *J Muscle Res Cell Motil* 26: 199–212, 2005.
- Fukuda N, Kajiwara H, Ishiwata S, Kurihara S. Effects of MgADP on length dependence of tension generation in skinned rat cardiac muscle. *Circ Res* 86: E1–E6, 2000.
- Godt RE, Lindley BD. Influence of temperature upon contractile activation and isometric force production in mechanically skinned muscle fibers of the frog. *J Gen Physiol* 80: 279–297, 1982.
- Granzier HL, Irving TC. Passive tension in cardiac muscle: contribution of collagen, titin, microtubules, and intermediate filaments. *Biophys J* 68: 1027–1044, 1995.
- Greenberg MJ, Shuman H, Ostap EM. Inherent force-dependent properties of β -cardiac myosin contribute to the force-velocity relationship of cardiac muscle. *Biophys J* 107: L41–L44, 2014.
- Hancock WO, Martyn DA, Huntsman LL. Ca^{2+} and segment length dependence of isometric force kinetics in intact ferret cardiac muscle. *Circ Res* 73: 603–611, 1993.
- Hanft LM, Greaser ML, McDonald KS. Titin-mediated control of cardiac myofibrillar function. *Arch Biochem Biophys* 552–553: 83–91, 2014.
- Hanft LM, McDonald KS. Length dependence of force generation exhibit similarities between rat cardiac myocytes and skeletal muscle fibres. *J Physiol* 588: 2891–2903, 2010.
- Hibberd MG, Jewell BR. Calcium- and length-dependent force production in rat ventricular muscle. *J Physiol* 329: 527–540, 1982.
- Hofmann PA, Fuchs F. Effect of length and cross-bridge attachment on Ca^{2+} binding to cardiac troponin C. *Am J Physiol Cell Physiol* 253: C90–C96, 1987.
- Hofmann PA, Fuchs F. Evidence for a force-dependent component of calcium binding to cardiac troponin c. *Am J Physiol Cell Physiol* 253: C541–C546, 1987.
- Huxley AF. Muscle structure and theories of contraction. *Prog Biophys Biophys Chem* 7: 255–318, 1957.
- Huxley AF, Simmons RM. Proposed mechanism of force generation in striated muscle. *Nature* 233: 533–538, 1971.
- Irving TC, Konhilas J, Perry D, Fischetti R, de Tombe PP. Myofilament lattice spacing as a function of sarcomere length in isolated rat myocardium. *Am J Physiol Heart Circ Physiol* 279: H2568–H2573, 2000.
- Kawai M, Halvorson H. Two step mechanism of phosphate release and the mechanisms of force generation in chemically skinned fibers of rabbit psoas muscle. *Biophys J* 59: 329–342, 1991.
- Kawai M, Schulman MI. Crossbridge kinetics in chemically skinned rabbit psoas fibres when the actin-myosin lattice spacing is altered by dextran T-500. *J Muscle Res Cell Motil* 6: 313–332, 1985.
- Kaya M, Higuchi H. Nonlinear elasticity and an 8-nm working stroke of single myosin molecules in myofilaments. *Science* 329: 686–689, 2010.
- Kobirumaki-Shimozawa F, Inoue T, Shintani SA, Oyama K, Terui T, Minamisawa S, Ishiwata S, Fukuda N. Cardiac thin filament regulation and the Frank-Starling mechanism. *J Physiol Sci* 64: 221–232, 2014.
- Konhilas JP, Irving TC, de Tombe PP. Frank-Starling law of the heart and the cellular mechanisms of length-dependent activation. *Pflügers Arch* 445: 305–310, 2002.
- Konhilas JP, Irving TC, de Tombe PP. Myofilament calcium sensitivity in skinned rat cardiac trabeculae: role of interfilament spacing. *Circ Res* 90: 59–65, 2002.
- Korte FS, Feest ER, Razumova MV, Tu AY, Regnier M. Enhanced Ca^{2+} binding of cardiac troponin reduces sarcomere length dependence of contractile activation independently of strong crossbridges. *Am J Physiol Heart Circ Physiol* 303: H863–H870, 2012.
- Korte FS, McDonald KS. Sarcomere length dependence of rat skinned cardiac myocyte mechanical properties: dependence on myosin heavy chain. *J Physiol* 581: 725–739, 2007.
- Krasner B, Maughan DW. The relationship between ATP hydrolysis and active force in compressed and swollen skinned muscle fibers of the rabbit. *Pflügers Arch* 400: 160–165, 1984.
- Li KL, Rieck D, Solaro RJ, Dong W. In situ time-resolved FRET reveals effects of sarcomere length on cardiac thin-filament activation. *Biophys J* 107: 682–693, 2014.
- Lymn RW, Taylor EW. Mechanism of adenosine triphosphate hydrolysis by actomyosin. *Biochemistry* 10: 4617–4624, 1971.
- Matsubara I, Millman BM. X-ray diffraction patterns from mammalian heart muscle. *J Mol Biol* 82: 527–536, 1974.
- McDonald KS, Moss RL. Osmotic compression of single cardiac myocytes eliminates the reduction in Ca^{2+} sensitivity of tension at short sarcomere length. *Circ Res* 77: 199–205, 1995.
- McKillop DF, Geeves MA. Regulation of the interaction between actin and myosin subfragment 1: evidence for three states of the thin filament. *Biophys J* 65: 693–701, 1993.
- Metzger JM. Myosin binding-induced cooperative activation of the thin filament in cardiac myocytes and skeletal muscle fibers. *Biophys J* 68: 1430–1442, 1995.
- Milani-Nejad N, Xu Y, Davis JP, Campbell KS, Janssen PML. Effect of muscle length on cross-bridge kinetics in intact cardiac trabeculae at body temperature. *J Gen Physiol* 141: 133–139, 2013.
- Mulieri LA, Barnes WD, Leavett BJ, Ittleman F, LeWinter MM, Alpert NR, Maughan DW. Alterations of myocardial dynamic stiffness implicating abnormal crossbridge function in human mitral regurgitation heart failure. *Circ Res* 90: 66–72, 2002.
- Nowak G, Pena JJR, Urboniene D, Geenen DL, Solaro RJ, Wolska BM. Correlations between alterations in length-dependent Ca^{2+} activation of cardiac myofilaments and the end-systolic pressure-volume relation. *J Muscle Res Cell Motil* 28: 415–419, 2007.

50. **Palmer BM, Suzuki T, Wang Y, Barnes WD, Miller MS, Maughan DW.** Two-state model of acto-myosin attachment-detachment predicts C-process of sinusoidal analysis. *Biophys J* 93: 760–769, 2007.
51. **Palmer BM, Tanner BCW, Toth MJ, Miller MS.** An inverse power-law distribution of molecular bond lifetimes predicts fractional derivative viscoelasticity in biological tissue. *Biophys J* 104: 2540–2552, 2013.
52. **Pate E, Cooke R.** A model of crossbridge action: the effects of ATP, ADP, and Pi. *J Muscle Res Cell Motil* 10: 181–196, 1989.
53. **Patel JR, Pleitner JM, Moss RL, Greaser ML.** Magnitude of length-dependent changes in contractile properties varies with titin isoform in rat ventricles. *Am J Physiol Heart Circ Physiol* 302: H697–H708, 2012.
54. **Reconditi M, Linari M, Lucii L, Stewart A, Sun YB, Boesecke P, Narayanan T, Fischetti RF, Irving T, Piazzesi G, ML, Lombardi V.** The myosin motor in muscle generates a smaller and slower working stroke at higher load. *Nature* 428: 578–581, 2004.
55. **Siemankowski RF, Wiseman MO, White HD.** ADP dissociation from actomyosin subfragment 1 is sufficiently slow to limit the unloaded shortening velocity in vertebrate muscle. *Proc Natl Acad Sci USA* 82: 658–662, 1985.
56. **Smith L, Tainter C, Regnier M, Martyn DA.** Cooperative cross-bridge activation of thin filaments contributes to the Frank-Starling mechanism in cardiac muscle. *Biophys J* 96: 3692–3702, 2009.
57. **Stelzer JE, Moss RL.** Contributions of stretch activation to length-dependent contraction in murine myocardium. *J Gen Physiol* 128: 461–471, 2006.
58. **Tanner BCW, Daniel TL, Regnier M.** Sarcomere lattice geometry influences cooperative myosin binding in muscle. *PLoS Comput Biol* 3: e115, 2007.
59. **Tanner BCW, Farman GP, Irving TC, Maughan DW, Palmer BM, Miller MS.** Thick-to-thin filament surface distance modulates cross-bridge kinetics in *Drosophila* flight muscle. *Biophys J* 103: 1275–1284, 2012.
60. **Tanner BCW, Wang Y, Maughan DW, Palmer BM.** Measuring myosin cross-bridge attachment time in activated muscle fibers using stochastic vs. sinusoidal length perturbation analysis. *J Appl Physiol* 110: 1101–1108, 2011.
61. **Tanner BCW, Wang Y, Robbins J, Palmer BM.** Kinetics of cardiac myosin isoforms in mouse myocardium are affected differently by presence of myosin binding protein-C. *J Muscle Res Cell Motil* 35: 267–278, 2014.
62. **ter Keurs HE, Bucx JJ, de Tombe PP, Backx P, Iwazumi T.** The effects of sarcomere length and Ca^{++} on force and velocity of shortening in cardiac muscle. *Adv Exp Med Biol* 226: 581–593, 1988.
63. **Tyska MJ, Warshaw DM.** The myosin power stroke. *Cell Motil Cytoskeleton* 51: 1–15, 2002.
64. **Wang Y, Tanner BCW, Lombardo AT, Tremble SM, Maughan DW, Vanburen P, Lewinter MM, Robbins J, Palmer BM.** Cardiac myosin isoforms exhibit differential rates of MgADP release and MgATP binding detected by myocardial viscoelasticity. *J Mol Cell Cardiol* 54: 1–8, 2013.
65. **Wang YP, Fuchs F.** Length, force, and Ca^{2+} -troponin C affinity in cardiac and slow skeletal muscle. *Am J Physiol Cell Physiol* 266: C1077–C1082, 1994.
66. **Wannenburg T, Janssen PM, Fan D, de Tombe PP.** The Frank-Starling mechanism is not mediated by changes in rate of cross-bridge detachment. *Am J Physiol Heart Circ Physiol* 273: H2428–H2435, 1997.
67. **Williams CD, Regnier M, Daniel TL.** Axial and radial forces of cross-bridges depend on lattice spacing. *PLoS Comput Biol* 6: e1001018, 2010.
68. **Zhao Y, Kawai M.** The effect of the lattice spacing change on cross-bridge kinetics in chemically skinned rabbit psoas muscle fibers. II. Elementary steps affected by the spacing change. *Biophys J* 64: 197–210, 1993.

